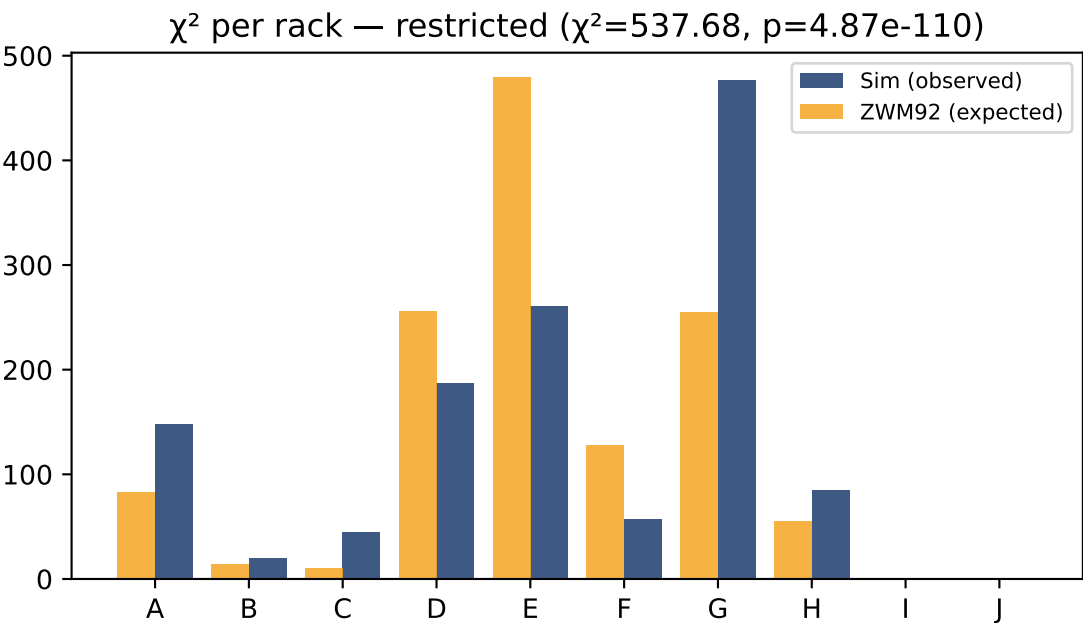


V&V5 — Operational Validity

χ^2 + paired-t + Holm-Wilcoxon



Paired t (log daily picks per material):
 $t = 20.303$, $p = 4.15e-76$, $n = 928$

Restricted $\chi^2 = 537.7$ (Cramer's $V = 0.216$ — small effect); pooled (Cochran OK): $\chi^2 = 537.7$

Replication volume calibration: PASS — Baseline (Actual SAP): simulated arrivals/day within 1.1% of ZWM92

Unpooled Cochran note: Cochran's rule violated: 2 of 10 cells have expected count < 5 — $(\text{'I'}, 0.04)$, $(\text{'J'}, 0.1)$.
Unpooled p-value i

CRN paired-by-replication t-tests vs Heuristic (N=20)

